ABSTRACT

This project presents an exploratory data analysis (EDA) of movie ratings to understand how audiences rate films and which genres and movies they prefer. The analysis uses the MovieLens ml-latest-small dataset, which contains 100,836 ratings given by 610 users to 9,724 movies between 1996 and 2018. The raw ratings were merged with movie titles and genres, cleaned, and extended with engineered fields such as release year, primary genre, rating year, rating category and a liked flag, giving a cleaned table of 100,836 rows and 14 columns with no duplicate rows.

Four visualizations were produced with Python (pandas and Matplotlib): the star-rating distribution, the top ten genres by number of ratings, the average rating by year, and the ten highest-rated movies with at least 50 ratings. The results show that 4.0 stars is the most common rating (26,818 ratings), the average rating is 3.50, Drama and Comedy receive the most ratings, and The Shawshank Redemption has the highest average rating (4.43) among movies with at least 50 ratings. The dataset records only ratings and not viewer details, so the conclusions describe rating behaviour and not the reasons behind it.

CHAPTER 1 – PROBLEM DEFINITION

1.1 Problem Statement

Movie platforms collect very large numbers of ratings, but the raw records do not directly show how viewers rate films or which genres and titles they prefer. The problem addressed in this project is to analyse a large movie-ratings dataset to reveal rating patterns, genre popularity, rating trends over time and the highest-rated movies, so that audience preferences can be described with evidence.

1.2 Analysis Focus

The central variable is rating, a star score from 0.5 to 5.0 in steps of 0.5 given by a user to a movie. Audience preference is studied by comparing this rating across genres, years and movies, together with the number of ratings each genre or movie receives, which indicates its popularity.

1.3 Dataset Variables

Variable	Description	Role
userId	Unique id of the user who gave the rating	Identifier
movieId	Unique id of the movie	Identifier
title	Movie title with the release year separated out	Descriptive
release_year	Year of movie release taken from the title	Derived field
genres	All genres of the movie separated by |	Categorical
primary_genre	First genre listed for the movie	Derived field
genre_count	Number of genres listed for the movie	Derived field
rating	Star rating from 0.5 to 5.0	Main analysis variable
rating_date	Date of the rating, converted from Unix timestamp	Time variable
rating_year	Year in which the rating was given	Time variable
movie_age_at_rating	Movie age (years) when it was rated	Derived field
release_decade	Decade of movie release	Derived field

Variable	Description	Role
rating_category	Low (below 3), Medium (3 to 3.5) or High (4 and above)	Derived field
liked	1 if the rating is 4 or more, otherwise 0	Derived field

1.4 Objectives

- Inspect the raw and cleaned ratings datasets for quality issues.
- Study how star ratings are distributed across the audience.
- Identify the genres that receive the most ratings.
- Examine how the average rating changes with the year in which ratings were given.
- Identify the highest-rated movies that have enough ratings to be reliable.
- Summarize audience preferences, limitations and future improvements.

CHAPTER 2 – DATA PREPARATION

2.1 Dataset Overview

Item	Details from the project workbook
Source	MovieLens ml-latest-small (GroupLens Research): ratings merged with movie titles and genres on movieId
Raw dataset	100,836 records, 6 columns (userId, movieId, rating, timestamp, title, genres)
Cleaned dataset	100,836 records, 14 columns (6 original and 8 engineered)
Users and movies	610 users and 9,724 movies
Rating period	1996 to 2018
Raw quality checks	No missing cells, no duplicate user and movie pairs, and no ratings outside 0.5 to 5.0
Items kept for review	13 titles with extra spaces, 18 rows without a release year and 47 rows with no genre listed
Cleaned missing values	18 each in release_year, movie_age_at_rating and release_decade; 0 in all other columns

2.2 Preparation Procedure

- Merge the ratings with movie titles and genres using movieId; no ratings were left unmatched.
- Trim extra spaces in titles and split the release year out of the title.
- Relabel "(no genres listed)" as "Unknown" and add primary_genre and genre_count.
- Convert the Unix timestamp to rating_date and rating_year; add movie_age_at_rating and release_decade.
- Add rating_category (Low, Medium, High) and the liked flag (rating of 4 or more).
- Upload the Cleaned Data sheet in Google Colab, convert rating and rating_year to numeric and drop rows without a rating.
- No rows were removed because none were invalid or duplicated; the 18 rows without a release year were kept and left out only of the decade analysis.

2.3 Data Preparation Output

The output printed by Colab for the dataset shape, missing values, duplicate rows and the number of unique movies and users is shown below.

```
Saving Movie_Rating_Analysis_Workbook.xlsx to Movie_Rating_Analysis_Workbook.xlsx
Dataset shape: (100836, 14)
\nMissing values:\n userId                 0
movieId                 0
title                   0
release_year           18
genres                  0
primary_genre           0
genre_count             0
rating                  0
rating_date             0
rating_year             0
movie_age_at_rating    18
release_decade         18
rating_category         0
liked                   0
dtype: int64
\nDuplicate rows: 0
\nUnique movies: 9724
Unique users: 610
```

CHAPTER 3 – EXPLORATORY DATA ANALYSIS

3.1 Rating Distribution

The number of ratings given at each star value from 0.5 to 5.0 is counted to show how generously or strictly the audience rates movies. A bar chart is used because the ratings are discrete values.

3.2 Genre Popularity

The genres field is split on the "|" symbol and expanded so that a movie is counted once in every genre it belongs to. The ten genres with the most ratings are then compared in a horizontal bar chart.

3.3 Rating Trend by Year

Ratings are grouped by the year in which they were given (rating_year) and the mean rating of each year is calculated and drawn as a line chart, to check whether audience rating behaviour changes over time.

3.4 Highest-Rated Movies

Movies are grouped by title and the average rating and number of ratings are calculated. Only movies with at least 50 ratings are kept so that a few enthusiastic ratings cannot produce a misleading list, and the ten best are ranked by average rating.

3.5 Analysis Code and Output

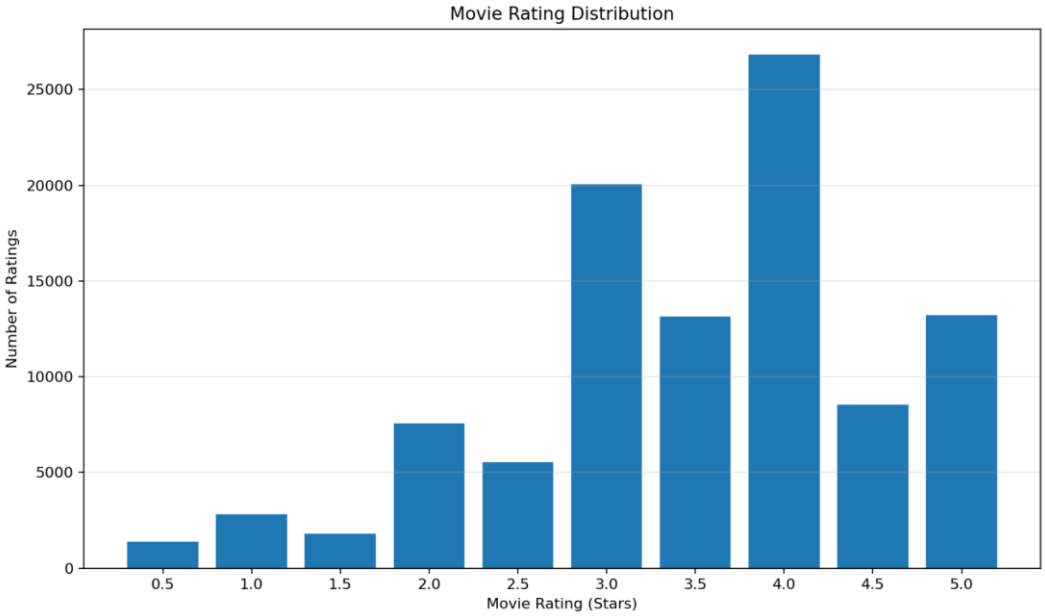
The complete runnable program is provided in the Appendix. The notebook, workbook, visualization graphs and presentation are available in the project repository: https://github.com/Zyrus-IC/Movie-Ratings-EDA. The notebook also saves the cleaned data as movie_ratings_cleaned_output.csv.

CHAPTER 4 – VISUALIZATION RESULTS

Visualization	Meaning
Rating distribution (bar chart)	Number of ratings at each star value
Top 10 genres (horizontal bar chart)	Genres that receive the most ratings
Average rating by year (line chart)	Mean rating in each year of rating
Top 10 highest-rated movies (horizontal bar chart)	Movies ranked by average rating, with at least 50 ratings

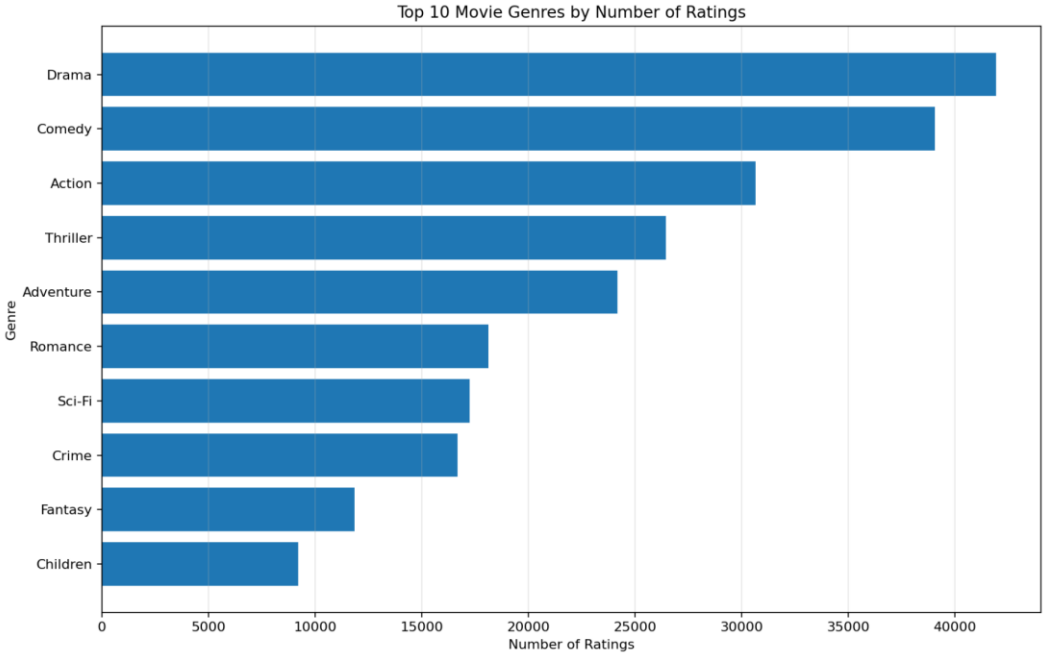
4.1 Rating Distribution

The most common rating is 4.0 stars with 26,818 ratings (26.6%), followed by 3.0 stars with 20,047 ratings and 5.0 stars with 13,211 ratings. Low ratings of 0.5 to 1.5 stars are rare. The average rating is 3.50 and the median is 3.5, and 48.2% of all ratings are 4 stars or higher, so viewers rate most movies favourably.



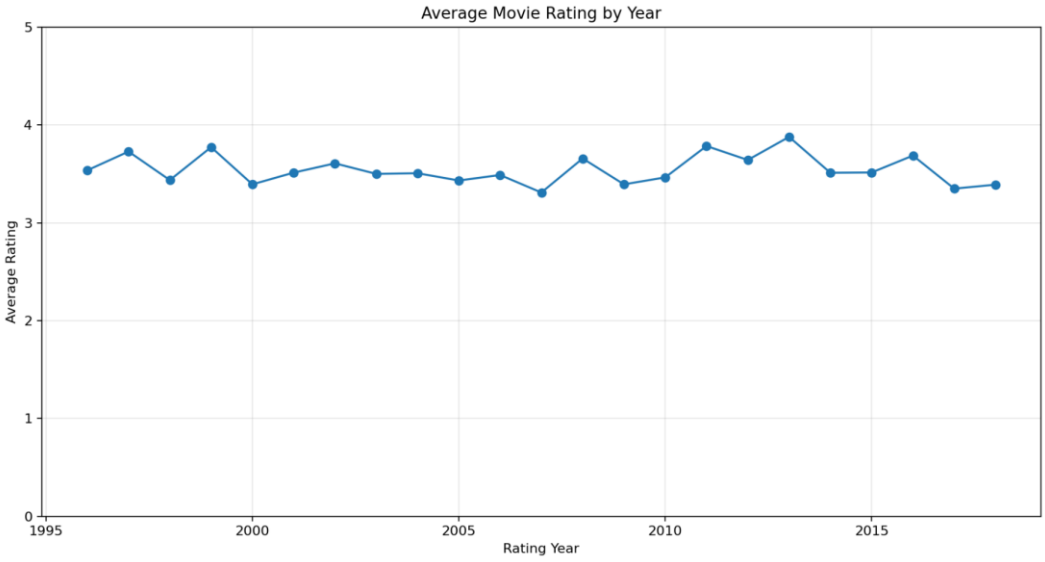
4.2 Genre Popularity

Drama (41,928 ratings) and Comedy (39,053) receive the most ratings, followed by Action (30,635), Thriller (26,452) and Adventure (24,161). Romance, Sci-Fi, Crime, Fantasy and Children complete the top ten.



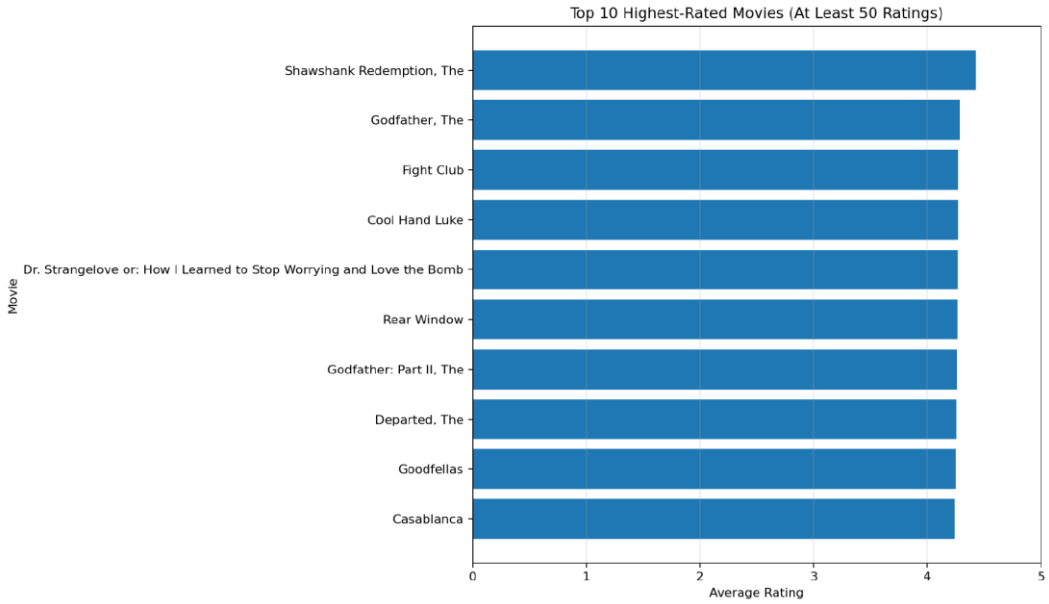
4.3 Average Rating by Year

The yearly average stays within a narrow band of about 3.3 to 3.9. The lowest yearly average is in 2007 (3.31) and the highest is in 2013 (3.88). The number of ratings per year differs greatly, from 507 in 1998 to 10,061 in 2000, so year-to-year changes should be read with care.



4.4 Top 10 Highest-Rated Movies

The Shawshank Redemption leads with an average rating of 4.43 from 317 ratings, followed by The Godfather (4.29), Fight Club (4.27), Cool Hand Luke (4.27) and Dr. Strangelove (4.27). All ten movies have an average rating above 4.2.



Note: A movie can belong to several genres, so the genre counts overlap and should not be added together.

CHAPTER 5 – AUDIENCE PREFERENCE COMPARISON

This chapter compares the audience groups found in the analysis: rating categories and genres. All counts are calculated from the cleaned dataset of 100,836 ratings.

Rating Category	Rule	Ratings	Share
Low	Below 3 stars	19,073	18.9%
Medium	3 to 3.5 stars	33,183	32.9%
High	4 stars and above	48,580	48.2%

Genre	Number of Ratings	Average Rating
Drama	41,928	3.66
Comedy	39,053	3.39
Action	30,635	3.45
Thriller	26,452	3.49
Adventure	24,161	3.51
Romance	18,124	3.51
Sci-Fi	17,243	3.46
Crime	16,681	3.66
Fantasy	11,834	3.49
Children	9,208	3.41

Popularity and approval are different measures. Drama and Comedy have the most ratings, but among the top ten genres Drama and Crime have the highest average rating (3.66) while Comedy (3.39) and Children (3.41) have the lowest. Across all genres in the dataset, Film-Noir (3.92, from only 870

ratings), War (3.81) and Documentary (3.80) have the highest averages and Horror has the lowest (3.26).

CHAPTER 6 – RESULT ANALYSIS

6.1 Key Findings

- The dataset contains 100,836 ratings from 610 users on 9,724 movies, given between 1996 and 2018, with no duplicate rows.
- The average rating is 3.50 (median 3.5); 4.0 stars is the most frequent rating and 48.2% of ratings are 4 stars or higher.
- Drama and Comedy are the most-rated genres, but Drama and Crime have higher average ratings than Comedy among the top ten.
- Yearly average ratings fluctuate between about 3.3 and 3.9 and show no clear long-term rise or fall.
- The ten highest-rated movies are well-known titles, mostly crime and drama films, led by The Shawshank Redemption (4.43).
- Users differ widely in activity: each user rated between 20 and 2,698 movies, with a median of 70.5 ratings per user.
- These patterns describe rating behaviour and do not establish the causes of audience preferences.

6.2 Audience Preference Insights

Audience preference was studied in two ways: popularity (the number of ratings) and approval (the average rating). Popularity points to Drama, Comedy and Action, while approval is highest for smaller genres such as Film-Noir, War and Documentary and for classic movies in the top-ten list. By release decade, movies released in the 1940s to 1970s have average ratings of about 3.8 to 3.9, compared with about 3.4 to 3.5 for movies from the 1980s to 2010s. This may be because viewers who choose to watch and rate older films are already interested in them, but the data cannot confirm the reason.

6.3 Limitations

- The dataset contains only ratings, titles and genres; it has no age, gender or location details, so preferences cannot be compared between viewer groups.
- The ratings come from 610 users who chose which movies to rate, so they may not represent all viewers.
- Genre counts overlap because one movie can belong to several genres.
- 18 ratings have no release year and 47 ratings have no genre listed; these were kept but are labelled or excluded in the affected analysis.
- Average ratings of movies with few ratings are unreliable, so a minimum of 50 ratings was used for the top-movie list.

6.4 Future Improvements

- Use a larger and more recent ratings dataset, with viewer details collected with consent.
- Analyse user segments such as generous, balanced and strict raters.
- Add genre-wise yearly trends and a deeper release-decade analysis.

- Build a movie recommendation system using the cleaned ratings data.
- Create an interactive dashboard to filter ratings by genre, year and rating value.

6.5 Real-World Applications

Rating analysis of this kind can help streaming platforms, theatres and producers understand which genres and titles audiences like, support content selection and recommendation systems, and track audience approval. This student project uses a public sample dataset and is intended for learning.

CONCLUSION

This report defines audience rating behaviour as the focus of study and describes the preparation of the MovieLens ratings data, the exploratory analysis and four visualizations covering rating distribution, genre popularity, yearly average rating and the highest-rated movies. Viewers most often give 4.0 stars, Drama and Comedy attract the most ratings, yearly averages stay close to 3.5, and classic crime and drama films top the approval ranking. The dataset has no viewer details and covers a limited group of users, so the findings describe rating patterns only and should not be treated as proof of why audiences prefer certain movies.

REFERENCES

- Project repository (notebook, workbook, visualization graphs and presentation): https://github.com/Zyrus-IC/Movie-Ratings-EDA
- MovieLens datasets, GroupLens Research: https://grouplens.org/datasets/movielens/
- Pandas documentation: https://pandas.pydata.org/docs/
- Matplotlib documentation: https://matplotlib.org/stable/
- Google Colab: https://colab.research.google.com/

APPENDIX – COMPLETE CODE

Notebook and workbook: https://github.com/Zyrus-IC/Movie-Ratings-EDA

```
# MOVIE RATING & AUDIENCE PREFERENCE ANALYSIS

# 1. Load Cleaned Dataset and Prepare Columns
import pandas as pd
import matplotlib.pyplot as plt
from google.colab import files

uploaded = files.upload()  # Select Movie_Rating_Analysis_Workbook.xlsx
file_name = next(iter(uploaded))

# Read the cleaned dataset from the workbook
df = pd.read_excel(file_name, sheet_name="Cleaned Data")

# Prepare columns
df["rating"] = pd.to_numeric(df["rating"], errors="coerce")
```

```
df["rating_year"] = pd.to_numeric(df["rating_year"], errors="coerce")
df = df.dropna(subset=["rating"])

print("Dataset shape:", df.shape)
print("\\nMissing values:\\n", df.isnull().sum())
print("\\nDuplicate rows:", df.duplicated().sum())
print("\\nUnique movies:", df["movieId"].nunique())
print("Unique users:", df["userId"].nunique())

# 2. Rating Distribution
rating_counts = df["rating"].value_counts().sort_index()
plt.figure(figsize=(10, 6))
plt.bar(rating_counts.index.astype(str), rating_counts.values)
plt.title("Movie Rating Distribution")
plt.xlabel("Movie Rating (Stars)")
plt.ylabel("Number of Ratings")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()

# 3. Top 10 Genres by Number of Ratings
genre_df = df[["genres"]].dropna().copy()
genre_df["genres"] = genre_df["genres"].astype(str).str.split("|")
genre_counts = (
    genre_df.explode("genres")["genres"]
    .value_counts()
    .head(10)
    .sort_values()
)
plt.figure(figsize=(11, 7))
plt.barh(genre_counts.index, genre_counts.values)
plt.title("Top 10 Movie Genres by Number of Ratings")
plt.xlabel("Number of Ratings")
plt.ylabel("Genre")
plt.grid(axis="x", alpha=0.3)
plt.tight_layout()
plt.show()


# 4. Average Rating by Year
yearly = (
    df.dropna(subset=["rating_year"])
    .groupby("rating_year")["rating"]
    .mean()
)
plt.figure(figsize=(11, 6))
plt.plot(yearly.index.astype(int), yearly.values, marker="o")
plt.title("Average Movie Rating by Year")
```

```
plt.xlabel("Rating Year")
plt.ylabel("Average Rating")
plt.ylim(0, 5)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

# 5. Top 10 Highest-Rated Movies (At Least 50 Ratings)
movie_stats = (
    df.groupby("title")
    .agg(avg_rating=("rating", "mean"),
         rating_count=("rating", "count"))
)
movie_stats = movie_stats[movie_stats["rating_count"] >= 50]
movie_stats = movie_stats.sort_values(
    ["avg_rating", "rating_count"], ascending=[False, False]
).head(10).sort_values("avg_rating")

plt.figure(figsize=(12, 7))
plt.barh(movie_stats.index, movie_stats["avg_rating"])
plt.title("Top 10 Highest-Rated Movies (At Least 50 Ratings)")
plt.xlabel("Average Rating")
plt.ylabel("Movie")
plt.xlim(0, 5)
plt.grid(axis="x", alpha=0.3)
plt.tight_layout()
plt.show()

# 6. Save Cleaned Output
df.to_csv("movie_ratings_cleaned_output.csv", index=False)
print("\\nMovie rating analysis completed successfully.")
```